The mixed-ratio weighted asymptotic

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Abstract

Fourth step of the mixed-ratio general-d monomial programme (Astra #15, route A). For arbitrary positive exponents ℓ_i ($i < d$) with minimum λ attained m times,

$$\frac{\int_{(0,1]^d} \prod_i t_i^{\ell_i - 1} e^{-\beta N \prod_i t_i}\, dt}{N^{-\lambda}(\log N)^{m-1}} \longrightarrow \frac{\Gamma(\lambda)\beta^{-\lambda}}{(m-1)!} \prod_{\ell_i > \lambda} \frac{1}{\ell_i - \lambda}$$

(`mixedBoxReal_tendsto`, equivalence form `mixedBoxReal_isEquivalent`). Zero `sorry`/`axiom`.

1 The things formalised

```
def multCount (  : Fin d →  ) (l :  ) :   :=   i, if   i = l then 1 else 0
def resFactor (  : Fin d →  ) (l :  ) :   :=   i, if   i = l then 1 else 1 / (  i - l)
def mixedConst ( ) (l   :  ) :   :=
  Gamma l *   ^ (-l) / ((multCount   l - 1).factorial :  ) * resFactor   l
theorem mixedBoxReal_tendsto (l  ) (hl : 0 < l) (h  : 0 <  ) :
      (d :  ) (  : Fin (d + 1) →  ), (  i, l     i) → (  i,   i = l) →
      Tendsto (fun N => mixedBoxReal (d + 1)     N
          / (N ^ (-l) * log N ^ (multCount   l - 1))) atTop (  (mixedConst   l  ))
theorem mixedBoxReal_isEquivalent (l  ) (hl) (h ) (d) ( ) (hmin) (hatt) :
    (fun N => mixedBoxReal (d + 1)     N)
      ~[atTop] powLog (mixedConst   l  ) l (multCount   l - 1)
```

2 Mathematics

Induction on the number of coordinates. If all exponents equal λ, the integral is the product density of unit 172 applied to $e^{-\beta N z}$ and the Gamma/log asymptotic of unit 173 gives the constant $\Gamma(\lambda)\beta^{-\lambda}/d!$ with log degree d (one less than the multiplicity $d+1$). Otherwise some $\ell_j > \lambda$; the swap $(0\,j)$ moves it to position 0 without changing the integral (unit 179), the multiplicity or the residue product (both are sums/products over all coordinates, invariant under `Equiv.sum_comp`/`Equiv.prod_comp`); the real recursion of unit 179 peels it, and the dominated-coordinate transfer lemma of unit 178 — applied to the lower-dimensional integral, which is measurable in its parameter and bounded by its mass — multiplies the constant by $1/(\ell_0 - \lambda)$ while preserving exponent and log degree. The minimal value remains attained in the tail because the attaining index is not 0.

3 Paper-facing meaning

This is the paper's zeta-function prediction for the (substituted) normal moment integral: exponent $\lambda = \min_i \ell_i$, logarithmic degree one less than the number of coordinates realising it, and the residue factor $\prod_{i \notin J} 1/(\ell_i - \lambda)$ of the Laurent coefficient a_{-m} (eq. `a_minus_m_explicit`, $b = 1$, before the Jacobian $1/(2k_i)$). The monomial assembly with $\ell_i = (h_i + 1)/(2k_i)$ is the next unit.

4 Lean notes

- A `def` whose body decides equality of reals (`if  i = l then ...`) must be `noncomputable`.
- `Finset.single_le_sum` needs the summand supplied explicitly (`(f := ...)`) when the goal is an `if`; the nonnegativity side goal is then `Nat.zero_le _`.
- `Subsingleton (Fin (0+1))` does not synthesise; use `Fin.ext (by omega)` for two elements of `Fin 1`.
- State multiplicities as `if` and residue products as `if` over `univ`: permutation invariance is then one lemma application and the first coordinate peels with `Fin.sum_univ_succ`/`Fin.prod_univ_succ`, avoiding `Finset.filter` cardinality bookkeeping.
- The transfer lemma takes the lower-dimensional integral as an anonymous function `fun M => mixedBoxReal ...`; the recursion lemma then matches after beta reduction without further rewriting.